# Thoughts for Statistical Mechanics

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# Chapter 1

# Ensemble

Why one does statistics? For fun? Maybe. But facing the problem that when particle number and hence number of Equations of Motion (EoM) approaches a limit, say, to order of $10^{23}$ in a commom macroscopic system, one would find it impossible to solve all these equations analytically at the same time, and add all the results together for macroscopic properties. Definitely this is because one has chosen the wrong method, or to say, the language. At this time, one should not stick to the naive mind of solving EoMs but rather, turn to some seemingly vaguely method to work it out, and that turns out to be statistics at this time.

How to understand statistics? Definitely, there are some mathematics inside. However, from a point of view in physics, statistics emerges because one is dealing with too many 'particles'; from the classical side, this is dealing with a high-dimensional point in Phase Space ($\Gamma$-space), whose movement is completely generated by the classical Hamiltonian $H(q,p)$. Since one is not sure — or one does not need statistics — about the exact orbital this point is in, one has to think about the possibility that the point may stay at every point — which is also impossible so one just treat it as some small area, which inherently shows some quantum nature, and the rest procedure for certain property is well known as some summation or integration. Then the problem falls on how to find such possibilities, which would be the main part of this chapter.

When statistics is suitable? Mathematically, possibility may be defined as frequency taking limit of infinite total events. Though such expression may fail sometimes, it should stand for physical processes. Translating this into physical language, it is easy to find that this equals the *Thermal Dynamical Limit*, i.e., when the total particle number $N$ and volume $V$ becomes infinity, while the number density $n = \frac{N}{V}$ is still finite, which is a physical condition. Under such consideration, the $\Gamma$ space is infinitely dimensional so the possible area the representative point could runs for and hence possible states become infinite. Note that because of uncertaintity relation by complementarity principle in quantum mechanics, minimal $\Gamma$ space area for one state is finite, then finite-dim $\Gamma$ space, together with closed orbitals generated by Hamiltonian, won't give countless states.

Since one is dealing with statistics, which is not exactly exact, to be honest, fluctuations are inevitable, which means that there might be 'small changes' to the system. Here, '' is used as we want to emphasis that when thinking about changes in statistics, they are more likely to be some virtual virations rather than real physical change. Because Statistical Physics here focuses on properties of system in equilibrium, inside which all states represent the same physical nature.

## 1.1 The Way Beginning: First Principle

*The first principle*, or *the postulate of equal probability*, claims that for one isolated macroscopic system in its thermodynamic equilibrium, its state is equally likely to be any state satisfying given macroscopic conditions.

Here some new concepts appear and one may find their meanings following the thought below. When could we confidently say that all states are equal? When the system is still changing? When the system is coupled to environment and is thus driven? No, because in both cases, some actions have been done on at least quite a part of microscopic particles and there must exist some preference states that the system like to stay, so to be confident enough, the system must be steady without any external factor, which is just *equilibrium*. In that case, one may identify the system as an *isolated* one. And clearly, the system has no energy/heat or particle exchange with the environment, and since it is steady, its volume

won't change as well. Such a system tagged with $(E, N, V)$ as invariant macroscopic conditions is said to composite *microcanonical ensemble*.

To find out the essence of statistics, one may start from a classical version. Given a system described by a Lagrangian $L(q, \dot{q})$, which does not depend on time apparently as we do want a steady state, who generates Euler-Lagrange EoMs by virations

$$\frac{\partial L}{\partial q_i} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} = 0, i = 1, \cdots, 3N.$$

Then by Legendre transformation, giving conjuagate momentum $p_i = \frac{\partial L}{\partial \dot{q}_i}$, Hamiltonian

$$H(q, p) = \dot{q} \cdot p - L,$$

with famous canonical EoMs

$$\frac{\partial H}{\partial p_i} = \dot{q}_i, \ \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i} = -\dot{p}_i, \tag{1.1}$$

which generates orbitals in $\Gamma$ space. In the sense that we live in 3-d Euclid space, focusing on $N$ particle system, our phase $\Gamma$ space $(q, p)$ is $6N$ dimensional. Clearly by its definition, each point in $\Gamma$ space represents a state of system, observing that we just treat $H$ as energy and since it does not contain time apparently,

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \sum_i \left(\frac{\partial H}{\partial q_i}\dot{q}_i + \frac{\partial H}{\partial p_i}\dot{p}_i\right) = \sum_i \left(-\dot{p}_i\dot{q}_i + \dot{q}_i\dot{p}_i\right) = 0,$$

i.e., once the system is in one state with energy $E$, the motion afterwards generated by (1.1) will force it stay on the curve, which lives on an equal-energy hypersurface defined by $H(q, p) = E$. And this just satisfies one of our macroscopic conditions. As we know clearly that partical number $N$ and volume $V$ do not change — the latter is because if it changes as in equilibrium, some work must be done and energy will change, too — we could say that such Hamiltonian just describes the isolated system we want.

Note that though many orbitals live on the one hypersurface, they do not run into each other since the canonical EoMs tell that the motion at each point is completely certain, let alone orbitals on different hypersurfaces. And now we may try to say something about statistics. As said before, we are not aiming at finding the exact state in $\Gamma$ space for the system, instead, we just want to know who can satisfy the simple $(E, N, V)$ condition, i.e., who cares which orbital the system is exactly evolving in? We know that all states on the hypersurface would make this work, and to make math easy, or to say due to quantum mechanical effect, we could loose the energy condition to $(E, E + \delta E)$, where $\delta E$ is some energy viration with no physical meaning — at most energy uncertaintity. But still, standing by the classical side, this means that we now have a shell with enomelous points of state inside — all satistying our macroscopic condition. And one thing certain is that we would like to count such points in some way, in order to utilize them. In the spirit of statistics, we introduce the states density function as

$$\rho(q, p, t)\mathrm{d}q\mathrm{d}p = \text{number of states, at t, in } \mathrm{d}q\mathrm{d}p,$$

also in spirit of quantum mechanics, each state would take place of $\sim h^{3N}$ in phase space. If we move along with one 'orbital' and consider 'certain size' of $\mathrm{d}q\mathrm{d}p$, we would say since volume is certain, number of states won't change, which leads us to famous Liouville's theorem,

$$0 = \frac{\mathrm{d}\rho}{\mathrm{d}t} = \frac{\partial \rho}{\partial t} + \sum_i \left(\frac{\partial \rho}{\partial q_i}\dot{q}_i + \frac{\partial \rho}{\partial p_i}\dot{p}_i\right) = \frac{\partial \rho}{\partial t} + \sum_i \left(\frac{\partial \rho}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial \rho}{\partial p_i}\frac{\partial H}{\partial q_i}\right). \tag{1.2}$$

For the steady state we are interested in, $\rho$ won't depend on time explicitly so

$$\frac{\partial \rho}{\partial t} = 0,$$

which in turn hints that $\rho = \rho(H)$ so that $\{\rho, H\} = 0$, but this is only one possible case.

Now, the first principle just claims that $\rho$ maintains not only with time, but also with phase space we are interested in, i.e.

$$\rho = \begin{cases} \text{Const}, \ E < H < E + \delta E \\ 0, \ \text{otherwise} \end{cases}, \tag{1.3}$$

which has already be hinted in the discussion before (1.2), as long as one admits that a state would take certain quantum volume, whose correctness is based on quantum probability.

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One most important thing one should keep in mind is that this postulate is just suitable for the so-called *microscopic ensemble*, with macroscopic condition $(E, N, V)$. Since just as we claimed before, external driven could have influence on the preference of the system. Here one word 'ensemble' appears, which is a statistical description for our system. Since we do not care — and we can not/not able to observe precisely — which state the system is in, we have to assume that the system may stay at any state between $(E, E + \delta E)$, and with this thought, one may natually think that for certain property we want to obtain, we have to do some average over all of those states, which is *ensemble average*, say that we have an observable $A$, then

$$\langle A \rangle = \frac{\int \mathrm{d}q\mathrm{d}p A(q,p)\rho(q,p)}{\int \mathrm{d}q\mathrm{d}p\rho(q,p)}, \tag{1.4}$$

which is suitable for all systems. In this sense, an ensemble is a collection of all possible states of given system, or to say, many copies satisfying the same macroscopic conditions.

Now, we would like to know one collective behavior of this ensemble, that is, how much derivation is the average value to its *most probable value*. And we may answer this by showing that the variation or the mean square fluctuation is small compared to the average, i.e.,

$$\frac{\langle f^2 \rangle - \langle f \rangle^2}{\langle f \rangle^2} \ll 1,$$

and we would find it later that this condition is satisfied in all physical cases. This property matters because in the former statement, we have already suggested some identity to all possible states, which in turn requires that any observable physical property for those states should be the same — or at least, the same in most cases, or they can not to be said in equal. Though we have subjectively fuzzed up those states, we still need a certain one for our system to sit in reality; if macroscopic properties of different state change dramatically, then there would be no utility for statistics — small changes in microscopic state would thoroughly change the system! Speaking of the identity, this reminds one of degenerate ground states for quantum systems, and all states are said to be indistinguishable, unless some other incompatiable observation is introduced. In that case, one natually do average on all possible ground states as they are exactly equal.

## 1.2 Microscopic Ensemble

This is a set of many copies of the system satisfying $(E, N, V)$ invariant macroscopic condition, but may be different at microscopic level.

Utilizing the homogeneity of density function, number of states occupied by this ensemble is:

$$\Gamma(E) = \frac{1}{N!h^{3N}} \int_{E<H<E+\delta E} \mathrm{d}^{3N}p\mathrm{d}^{3N}q, \tag{1.5}$$

which could be seen as

$$\Gamma(E) = \omega(E)\delta E = \frac{\partial \Sigma(E)}{\partial E}\delta E, \tag{1.6}$$

with phase space volume and density of states

$$\Sigma(E) = \frac{1}{N!h^{3N}} \int_{H<E} \mathrm{d}^{3N}p\mathrm{d}^{3N}q, \tag{1.7a}$$

$$\omega(E) = \frac{\partial \Sigma(E)}{\partial E}, \tag{1.7b}$$

where $h$ is phase space volume per state occupied and is found eventually to be *Planck's constant*; $N!$ here stands for another quantum mechanical effect named as *identical particles*, which claims that interchange two identical particles in one system would give at most a phase difference on the quantum state, and thus equal for classical counting, and this coefficent also has a formal name as *correct Boltzmann counting*. And in this sense we have chosen

$$\rho(q,p) = \frac{1}{N!h^{3N}}.$$

Now we are able to say something. We already know some macroscopic variables in thermodynamics, but in statistical mechanics, we only have quantities defined above, and we need to find a link. Define *entropy*

$$S = k \ln \Gamma(E), \tag{1.8}$$

where $k$ is found to be Boltzmann's constant. Clearly, entropy depends on the macroscopic condition observables, i.e. $S = S(E, N, V)$, which could also been by the definition. Entropy defined here is obviously an extensive quantity, since number of states obeys multiplication and gives entropy addition. Also note that mixing two systems would definitely gives energy re-allocation, and due to our equal probability statement, together with the most probable is equivalent to ensemble average idea, we could say that in the equilibrium end, if the mixed system would like to follow statistics and sit in the most probable state, it has to make the total number of states maximized, i.e., total entropy reaches its maximum, which just comes to *the second law of thermodynamics*. In that case, we could inversely thinking by deviding one system into two and say

$$\delta[\Gamma_1(E_1)\Gamma_2(E_2)] = \Gamma_2\frac{\partial\Gamma_1}{\partial E_1}\delta E_1 + \Gamma_1\frac{\partial\Gamma_2}{\partial E_2}\delta E_2 = 0,$$

with the constraint $\delta E_1 + \delta E_2 = 0$, one reaches

$$\left[\frac{\partial S_1(E_1)}{\partial E_1}\right]_{E_1=\overline{E}_1} = \left[\frac{\partial S_2(E_2)}{\partial E_2}\right]_{E_2=\overline{E}_2}, \tag{1.9}$$

which immediately gives us definition of temperature as *0th law of thermodynamics*

$$\frac{1}{T} = \frac{\partial S}{\partial E}. \tag{1.10}$$

Here $T$ is absolute temperature, as a physical one.

Now we claim that in thermodynamic limit, the below definitions of entropy are equivalent

$$S = k\ln\Sigma(E) = k\ln\Gamma(E) = k\ln\omega(E).$$

Thermodynamic limit here is important caused that these expressions in fact differ by order $O(\ln N)$, which is caused by the derivation of volume. Some may argue that $\delta E$ should be 0, but that is not a physical case, where we should not count continuously but concretely around the energy shell.

To see more familar physics,

$$\begin{aligned}\mathrm{d}S(E,V,N) &= \left(\frac{\partial S}{\partial E}\right)_{V,N}\mathrm{d}E + \left(\frac{\partial S}{\partial V}\right)_{E,N}\mathrm{d}V + \left(\frac{\partial S}{\partial N}\right)_{E,V}\mathrm{d}N \\ &= \frac{1}{T}\mathrm{d}E + \frac{p}{T}\mathrm{d}V - \frac{\mu}{T}\mathrm{d}N,\end{aligned}$$

with pressure and chemical potential

$$p = T\left(\frac{\partial S}{\partial V}\right)_{E,N},\ \mu = -T\left(\frac{\partial S}{\partial N}\right)_{E,V},$$

and this would give us *the first law og thermodynamics* by setting $U = E$

$$\mathrm{d}U = \mathrm{d}E = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}N, \tag{1.11}$$

with

$$T = \left(\frac{\partial U}{\partial S}\right)_{V,N},\ p = -\left(\frac{\partial U}{\partial V}\right)_{E,N},\ \mu = \left(\frac{\partial U}{\partial N}\right)_{E,V},$$

and const-volume heat capacity

$$C_V = \left(\frac{\partial U}{\partial T}\right)_{V,N} = T\left(\frac{\partial S}{\partial T}\right)_{V,N}.$$

**Equipartition Theorem**

Using ensemble average,

$$\left\langle q_i\frac{\partial H}{\partial q_j}\right\rangle = \frac{1}{\Gamma(E)}\frac{1}{N!h^{3N}}\int_{E<H<E+\delta E}\mathrm{d}p\mathrm{d}q\ q_i\frac{\partial H}{\partial q_j} = \frac{\delta E}{\Gamma(E)}\frac{1}{N!h^{3N}}\frac{\partial}{\partial E}\int_{H<E}\mathrm{d}p\mathrm{d}q\ q_i\frac{\partial H}{\partial q_j}$$

$$
\begin{aligned}
&= \frac{1}{\omega(E)}\frac{1}{N!h^{3N}}\frac{\partial}{\partial E}\int_{H<E}\mathrm{d}p\mathrm{d}q\left[\frac{\partial}{\partial q_j}(q_i H) - H\frac{\partial q_i}{\partial q_j}\right] \\
&= \frac{1}{\omega(E)}\frac{1}{N!h^{3N}}\frac{\partial}{\partial E}\int_{H<E}\mathrm{d}p\mathrm{d}q\left[\frac{\partial}{\partial q_j}[q_i(H-E)] + (E-H)\frac{\partial q_i}{\partial q_j}\right] \\
&= \frac{1}{\omega(E)}\frac{1}{N!h^{3N}}\left\{C\left[q_i(H-E)\right]\big|_{H=E} + \delta_{ij}\left[\int_{H<E}\mathrm{d}p\mathrm{d}q\frac{\partial}{\partial E}(E-H) + \frac{\partial}{\partial E}\left(\int_{H<E}\mathrm{d}p\mathrm{d}q\right)(E-H)\right]\right\} \\
&= \frac{1}{\omega(E)}\delta_{ij}\Sigma(E) = \delta_{ij}\frac{\Sigma(E)}{\partial\Sigma(E)/\partial E} = \delta_{ij}\left[\frac{\partial}{\partial E}\ln\Sigma(E)\right]^{-1} = \delta_{ij}\frac{k}{\partial S/\partial E} \\
&= \delta_{ij}kT, \\
\left\langle p_i\frac{\partial H}{\partial p_j}\right\rangle &= \delta_{ij}kT,
\end{aligned}
$$

as the *generalized equipartition theorem*.

Note that by canonical EoMs,

$$
\left\langle\sum_{i=1}^{3N} q_i\dot{p}_i\right\rangle = -\left\langle\sum_{i=1}^{3N} q_i\frac{\partial H}{\partial q_i}\right\rangle = -3NkT,
$$

as *virial theorem*.

For quadratic Hamiltonians like

$$
H = \sum_i (A_i p_i^2 + B_i q_i^2) = \frac{1}{2}\sum_i\left(p_i\frac{\partial H}{\partial p_i} + q_i\frac{\partial H}{\partial q_i}\right),
$$

we have *equipartition theorem*

$$
\langle H\rangle = \frac{1}{2}fkT,\ \ f = 2(\Sigma_i).
$$

We may also have

$$
C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{1}{2}fk,
$$

which only holds at classical limit.

## 1.3 Canonical Ensemble